AI ACADEMIC MENTOR

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Team Name	CSE4104-7C-T05
Section	7C
Project Title	AI Academic Mentor — Personalized Learning & Study Assistant
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TEAM INFORMATION

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Project Information

About the Application

The AI Academic Mentor is a full-stack web application designed to serve as a personalized learning and study assistant for students at NUBTK. The platform combines traditional academic tools (study scheduling, note management) with AI-powered capabilities (contextual tutoring, adaptive quiz generation, intelligent summarization) to create a comprehensive academic support system.

Core Features

- AI Chatbot Tutor: Conversational AI powered by Google Gemini API for subject-specific academic tutoring with session history
- Quiz Generator: AI-generated MCQ and short-answer quizzes with real-time scoring and performance analytics
- Note Summarizer: Paste lecture notes or upload documents for instant AI-powered summarization
- Study Planner: Weekly drag-and-drop schedule with deadline tracking and AI-generated study plans
- Progress Analytics: Topic-level mastery heatmaps, streak tracking, and score trend visualization
- Admin Panel: Full platform control with user management, announcements, and system analytics

Design Philosophy

The application employs a premium dark-themed interface with a deep navy colour palette. All screens were designed at 1440px desktop width with full mobile responsiveness at 390px. The design prioritizes clarity, academic utility, and minimal cognitive overhead for student users.

Application Feature Set:

- **AI Chatbot Tutor:** Conversational AI powered by Google Gemini API for subject-specific tutoring
- **Quiz Generator:** AI-generated MCQ and short-answer quizzes with real-time scoring and analytics
- **Note Summarizer:** Paste lecture notes or upload documents for instant AI-powered summarization
- **Study Planner:** Drag-and-drop weekly schedule with deadline tracking and priority labeling
- **Progress Analytics:** Topic-level mastery heatmaps, streak tracking, and score trend visualization
- **Admin Panel:** Full platform control with user management, announcements, and system analytics

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1. Desktop UI Wireframes & Screenshots (1440px)

Screen 1: Landing Page

The landing page serves as the application entry point for new and returning NUBTK students. It communicates the platform's five core features at a glance using a minimal, focused dark layout.

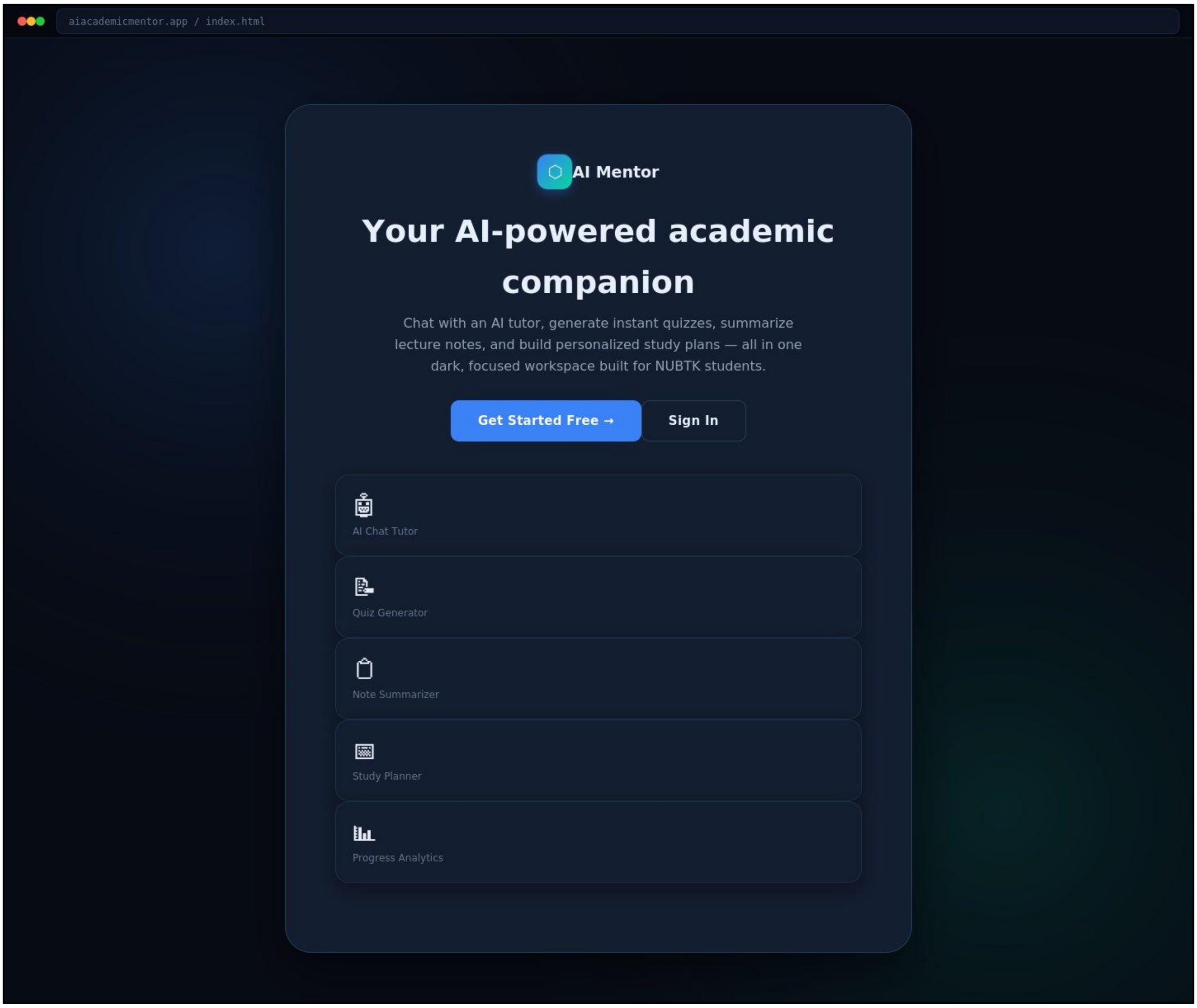

Figure: *Landing Page — Desktop View (1440px)*

Mobile Responsive View (390px):

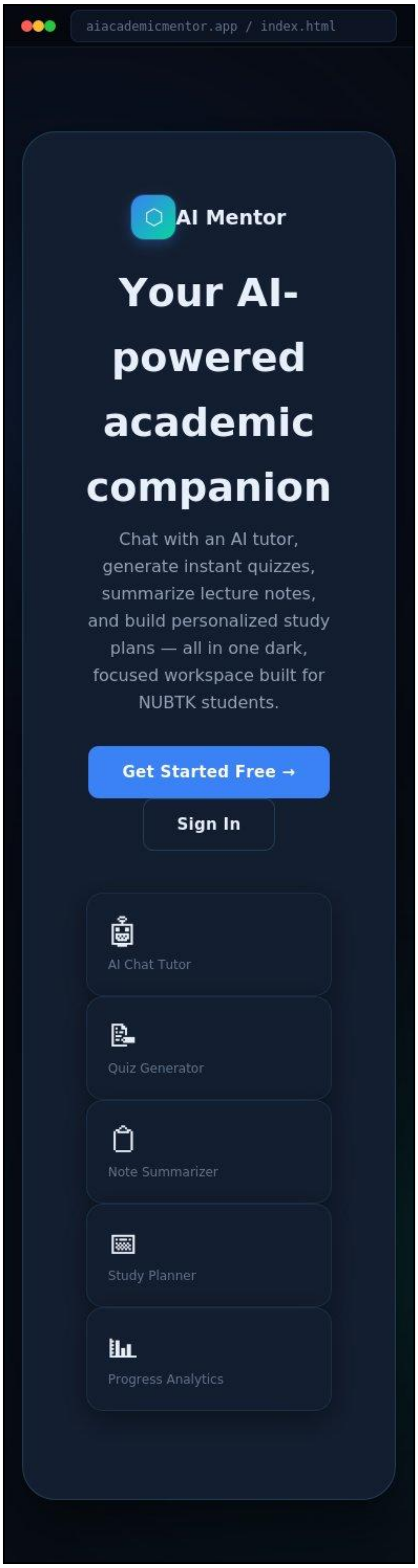

Figure: *Landing Page — Mobile View (390px)*

Key UI Elements:

- **Hero Section:** Full-viewport hero with radial gradient backgrounds and centered CTA cards
- **Feature Grid:** 5-column icon grid showcasing AI Chat, Quiz, Summarizer, Planner, Analytics
- **CTA Buttons:** Primary "Get Started Free" and secondary "Sign In" with hover lift animation
- **Logo Mark:** Hexagonal gradient mark with blue-to-teal gradient and glow shadow

Screen 2: Login Screen

Clean authentication interface with radial gradient ambient lighting. The card-based layout isolates focus on the form with no distracting navigation elements.

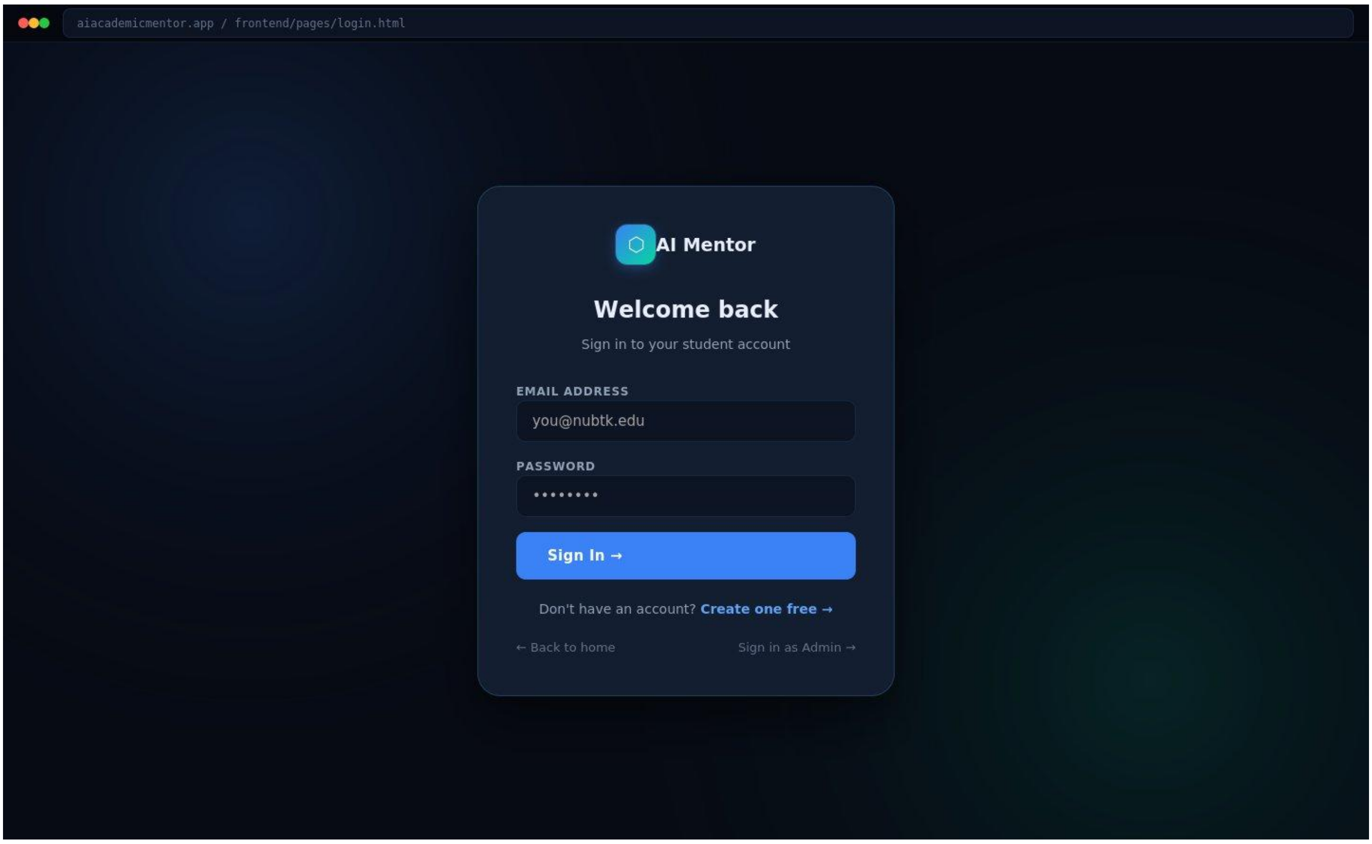

Figure: *Login Screen — Desktop View (1440px)*

Mobile Responsive View (390px):

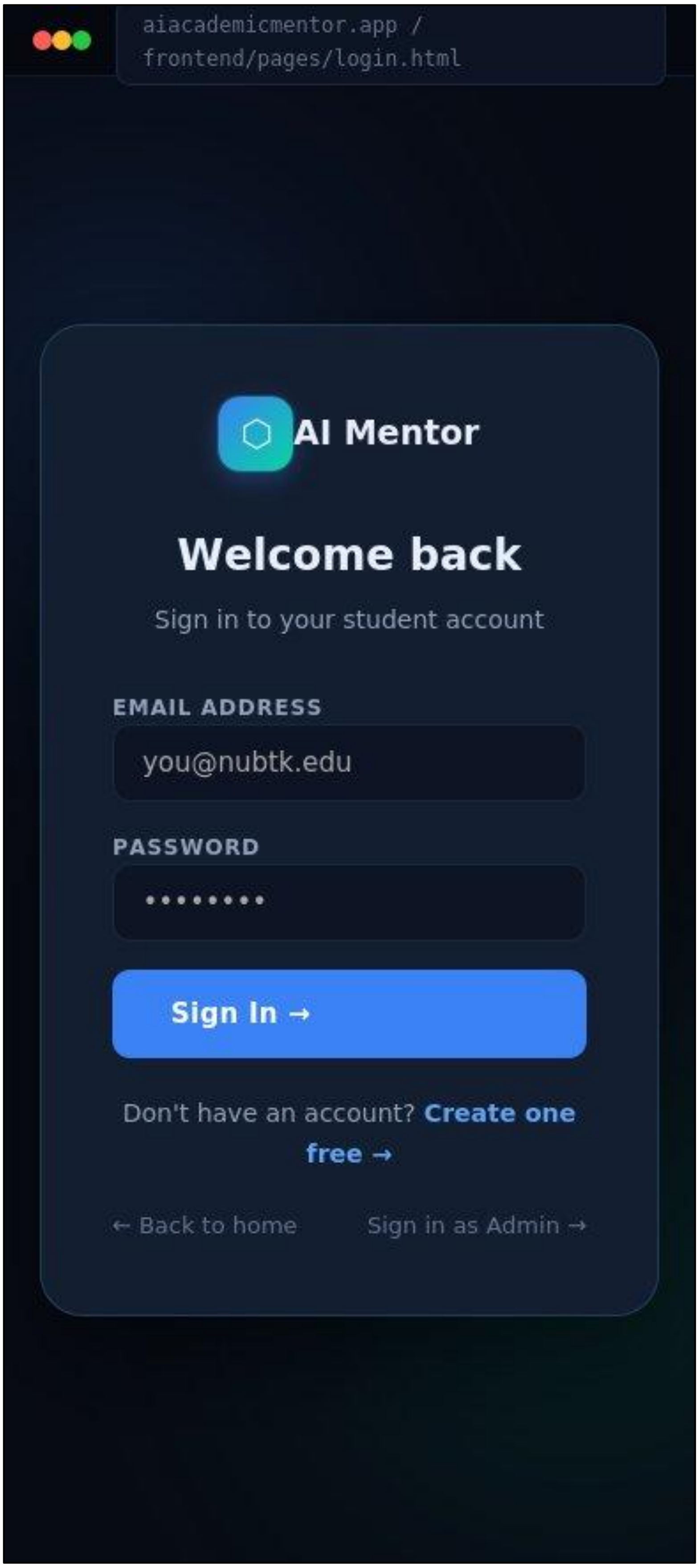

Figure: *Login Screen — Mobile View (390px)*

Key UI Elements:

- **Auth Card:** 440px centered card with glass-like border and radial glow background
- **Input Fields:** Dark bg-2 inputs with blue focus glow (0 0 0 3px primary-glow)
- **CTA:** Full-width primary Sign In button with → affordance
- **Navigation Links:** Back to home and Admin sign-in links with consistent text-3 styling

Screen 3: Registration Screen

Wider 560px card accommodates the more complex registration form with a two-column name row. All form elements follow the unified input system with consistent focus rings.

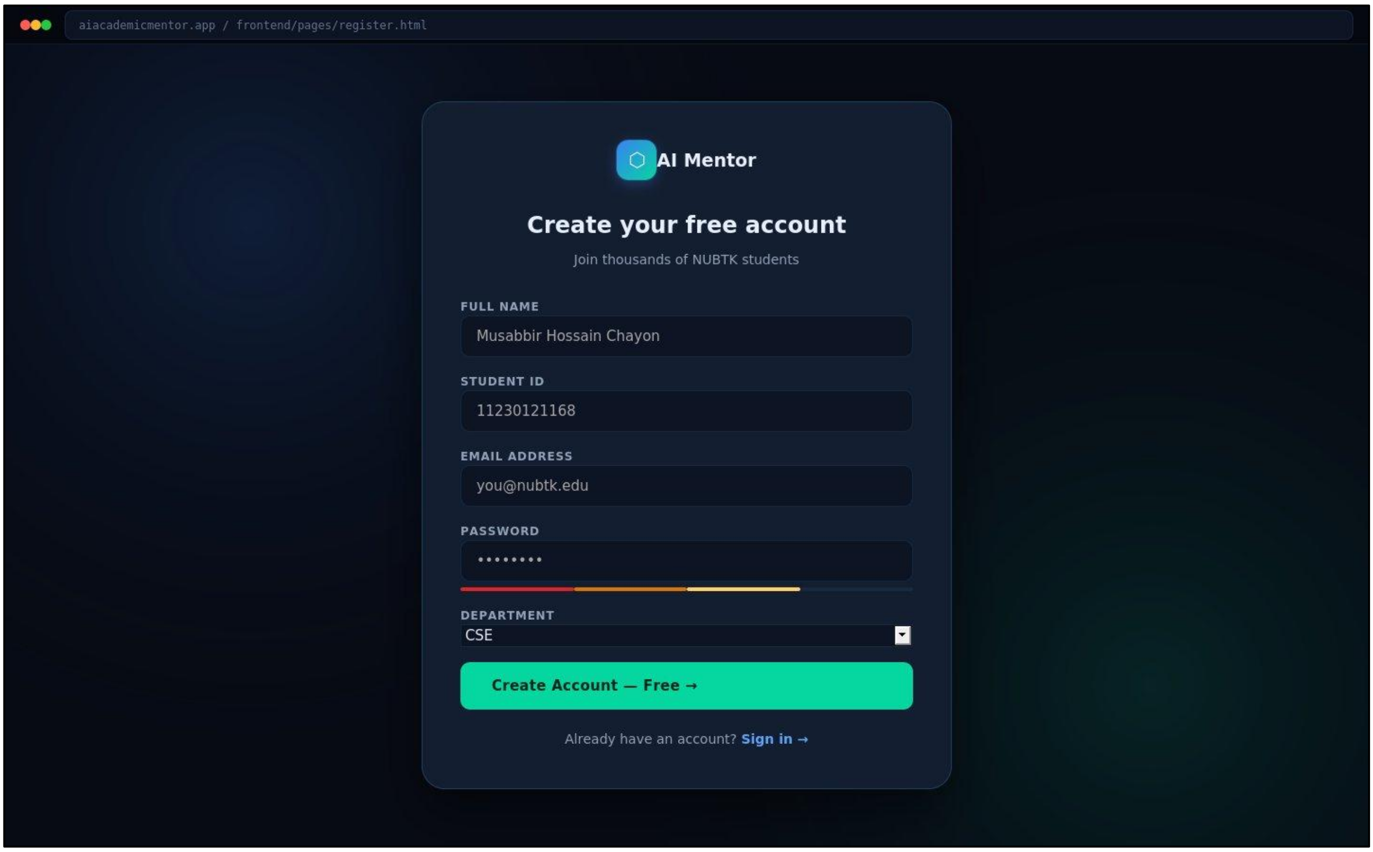

Figure: *Registration Screen — Desktop View (1440px)*

Mobile Responsive View (390px):

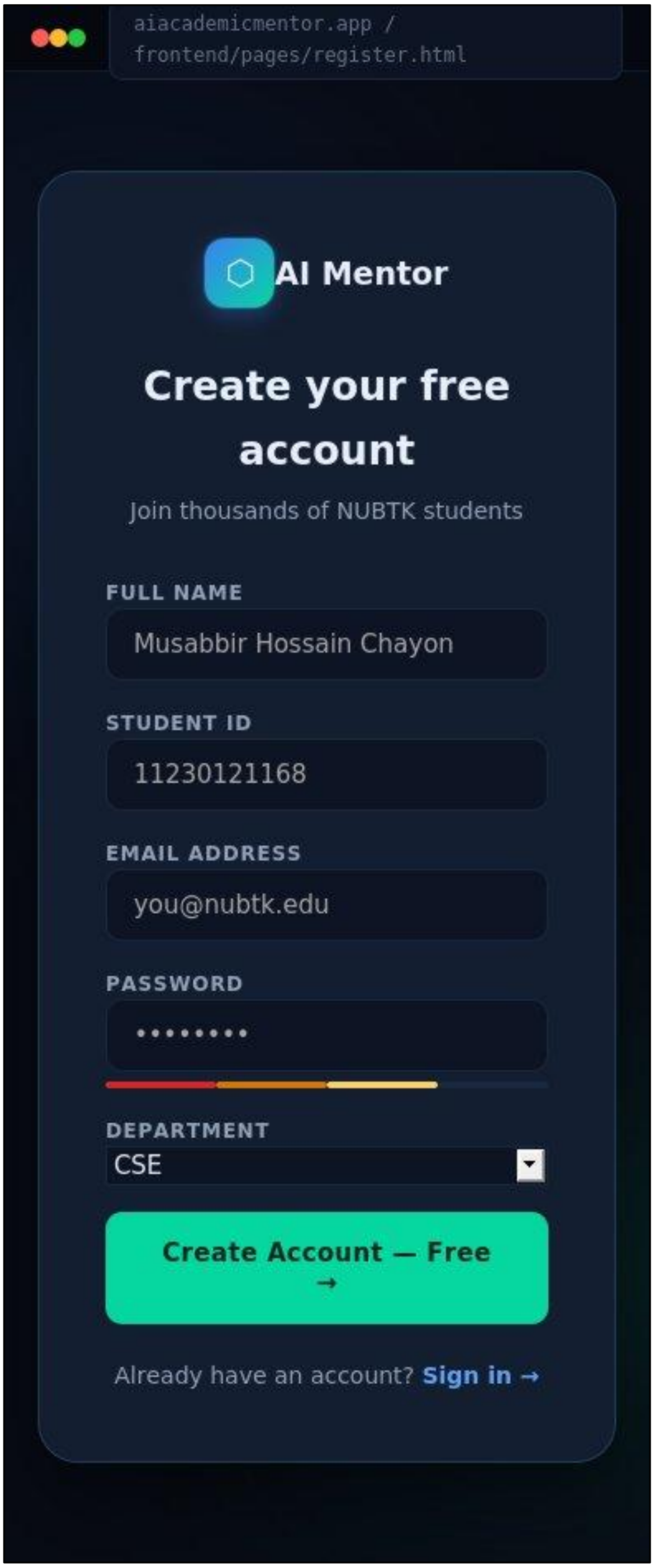

Figure: *Registration Screen — Mobile View (390px)*

Key UI Elements:

- **Two-Column Layout:** First Name / Last Name in responsive row-2 grid
- **Department Select:** Custom-styled dark select for CSE, EEE, BBA department choice
- **Student ID Field:** Dedicated field with NUBTK ID format placeholder
- **Role Toggle:** Inline Student / Admin role chip selector for account type

Screen 4: Dashboard

Central command hub. The dashboard presents all critical academic metrics and quick-access entry points to every feature in a scannable single-page layout.

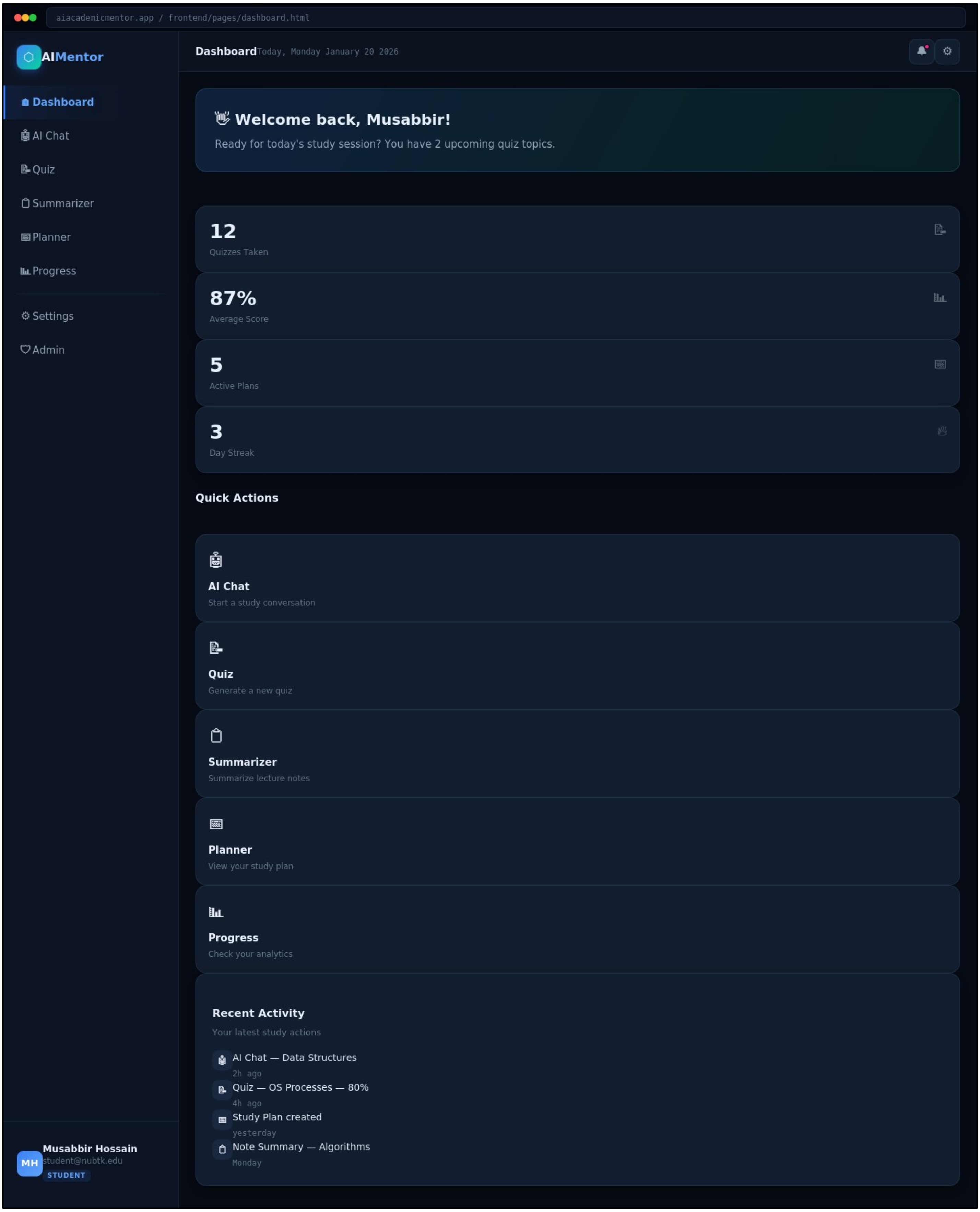

Figure: *Dashboard — Desktop View (1440px)*

Mobile Responsive View (390px):

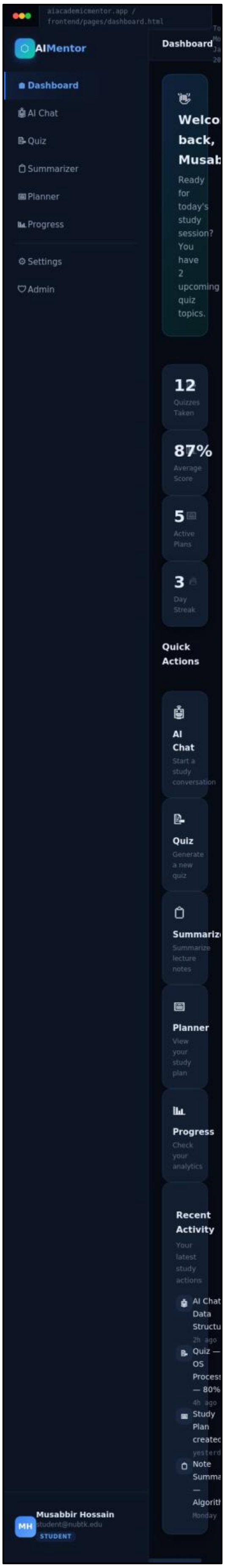

Figure: Dashboard — Mobile View (390px)

Key UI Elements:

- **Sidebar Navigation:** 260px sidebar with section labels, active state highlight, and role badge
- **Stats Row:** 5 metric cards — Active Plans, Quizzes Taken, Avg. Score, Streak, AI Sessions
- **Quick Actions:** Four gradient action cards for the four main AI features
- **Upcoming Tasks:** Priority-labeled task list with HIGH/MED/LOW urgency chips
- **Recent Quiz Results:** Score chip + subject label for last 3 quiz attempts
- **Streak Calendar:** Heatmap-style 7-day activity strip with active/inactive states

Screen 5: AI Chatbot Tutor

Full-viewport chat interface following familiar messaging conventions adapted for academic tutoring. The Gemini API powers contextual subject-specific responses.

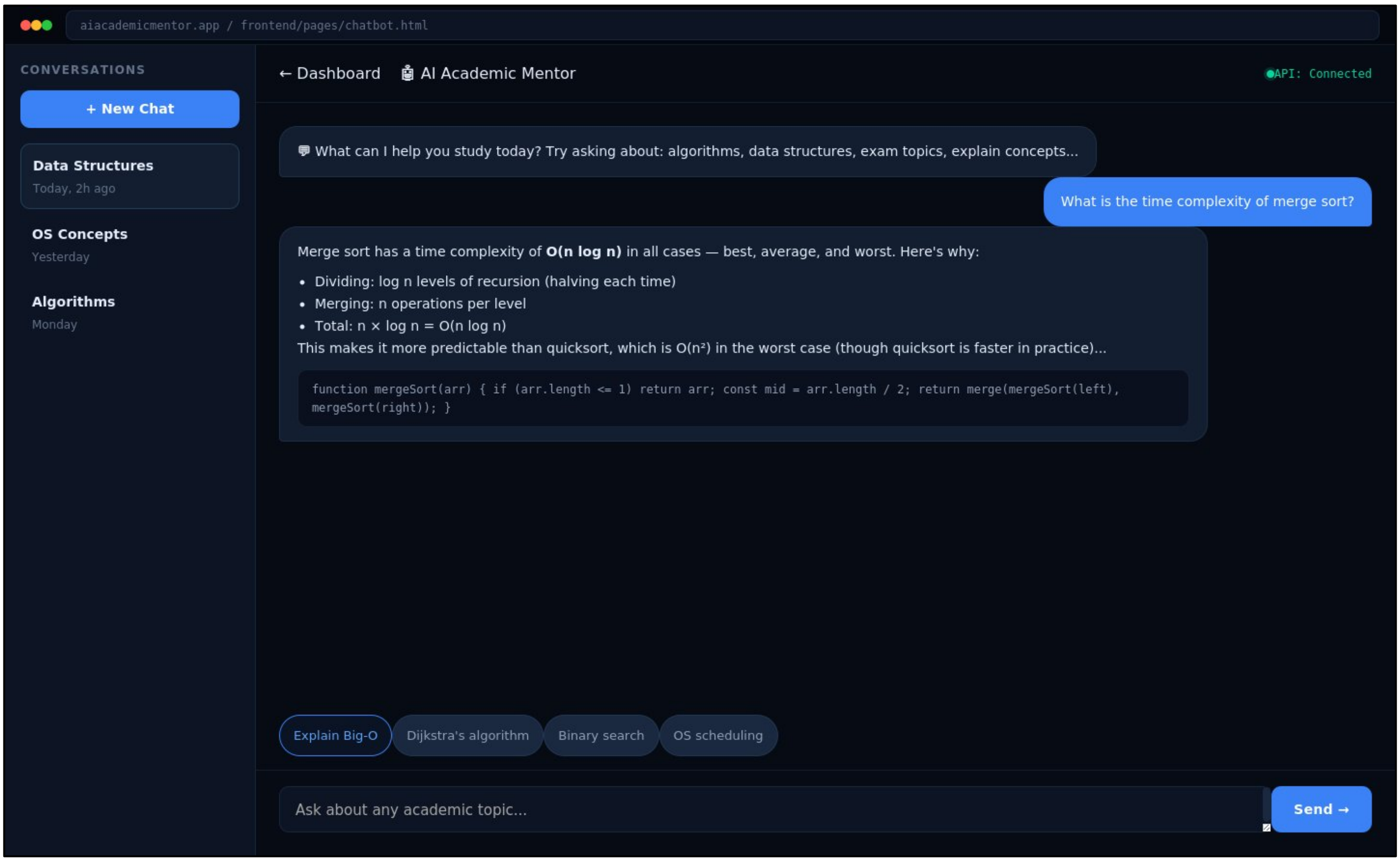

Figure: *AI Chatbot Tutor — Desktop View (1440px)*

Mobile Responsive View (390px):

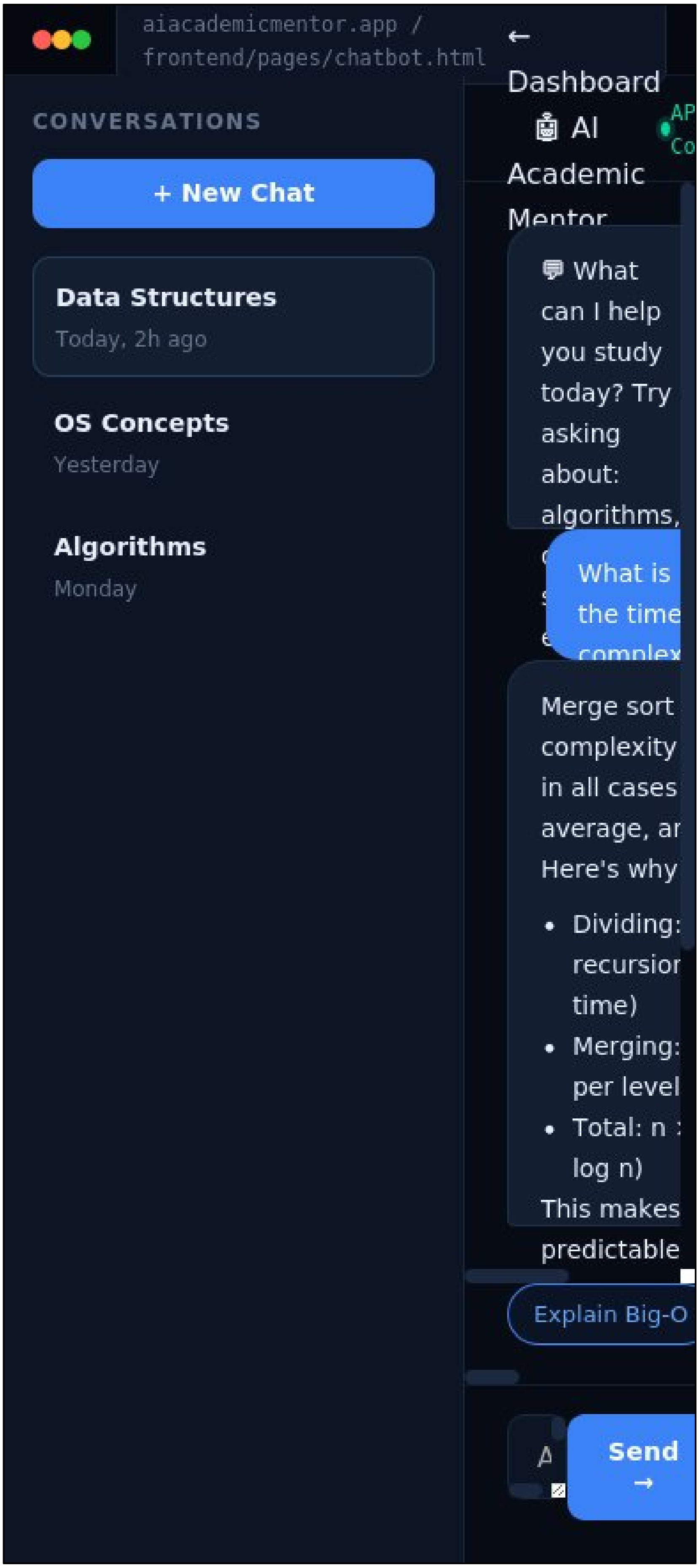

Figure: *AI Chatbot Tutor — Mobile View (390px)*

Key UI Elements:

- **Message Thread:** Alternating user (right-aligned, blue bubble) and AI (left, surface card) messages
- **AI Response Card:** Distinct surface-2 background with AI Mentor avatar initials
- **Input Bar:** Sticky bottom bar with textarea and Send button with shortcut hint
- **Suggested Prompts:** Quick-fire chip row for common student questions
- **Subject Selector:** Top dropdown to switch AI context between subjects

Screen 6: Quiz Generator

Two-state interface: configuration flow before generation, then question-by-question MCQ flow. The AI generates questions dynamically via the Gemini API based on the selected subject and difficulty.

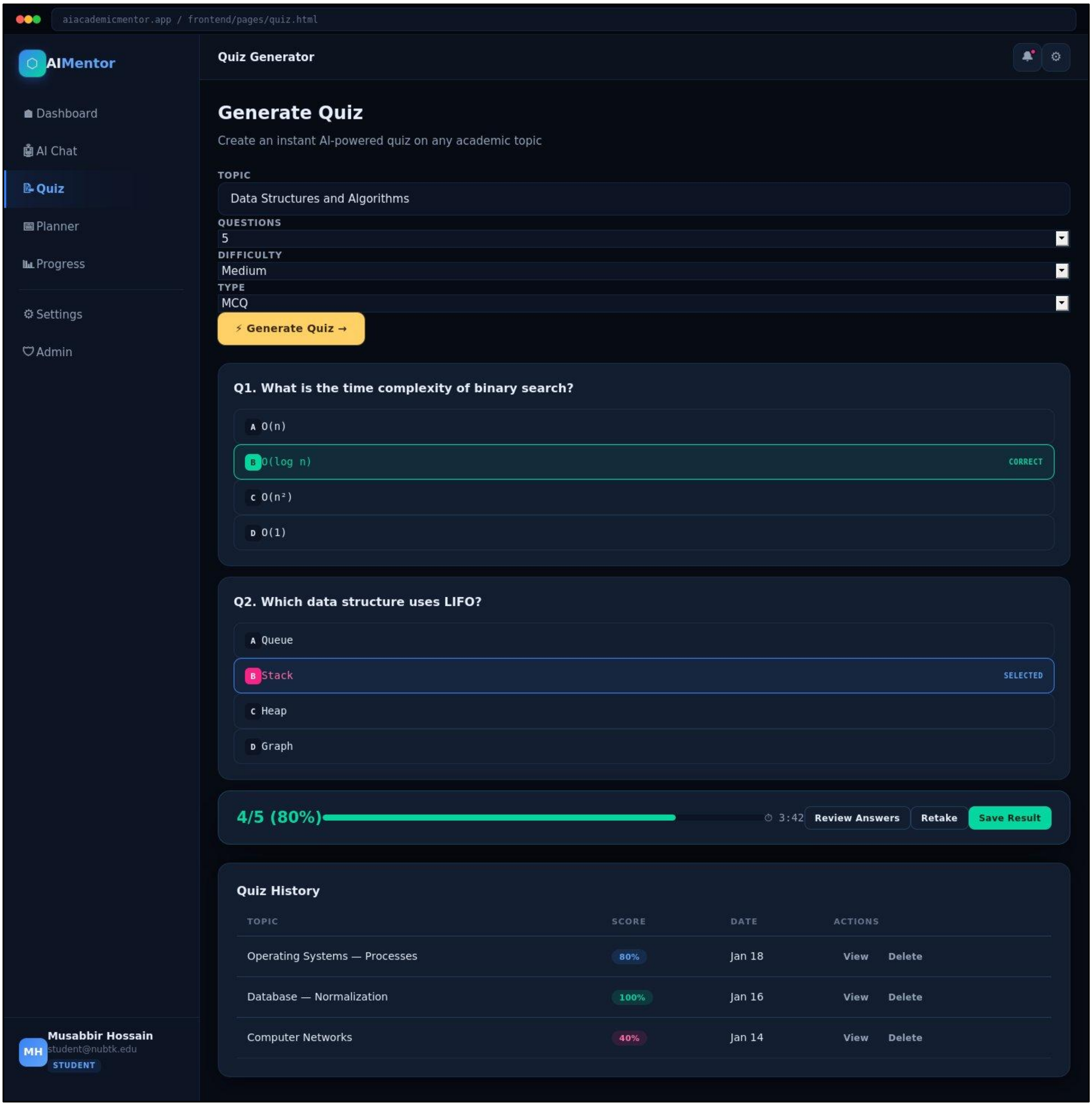

Figure: *Quiz Generator — Desktop View (1440px)*

Mobile Responsive View (390px):

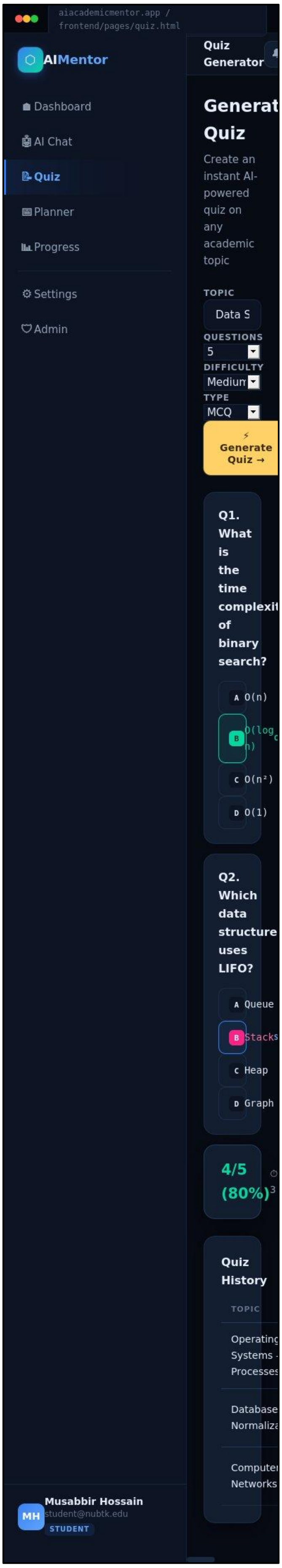

Figure: Quiz Generator — Mobile View (390px)

Key UI Elements:

- **Config Panel:** Subject, difficulty, question count, and format selectors before generation
- **Question Card:** Full-width MCQ card with A–D option buttons and active selection state
- **Progress Bar:** Thin accent-colored progress strip with current/total indicator
- **Score Summary:** Post-quiz results card with percentage ring, correct/wrong counts, and retry CTA
- **Timer Display:** JetBrains Mono countdown in the topbar during active quiz session

Screen 7: Note Summarizer

Two-panel layout: input on the left, AI output on the right (or stacked on mobile). Students paste or upload lecture content and receive instant structured summaries powered by Gemini.

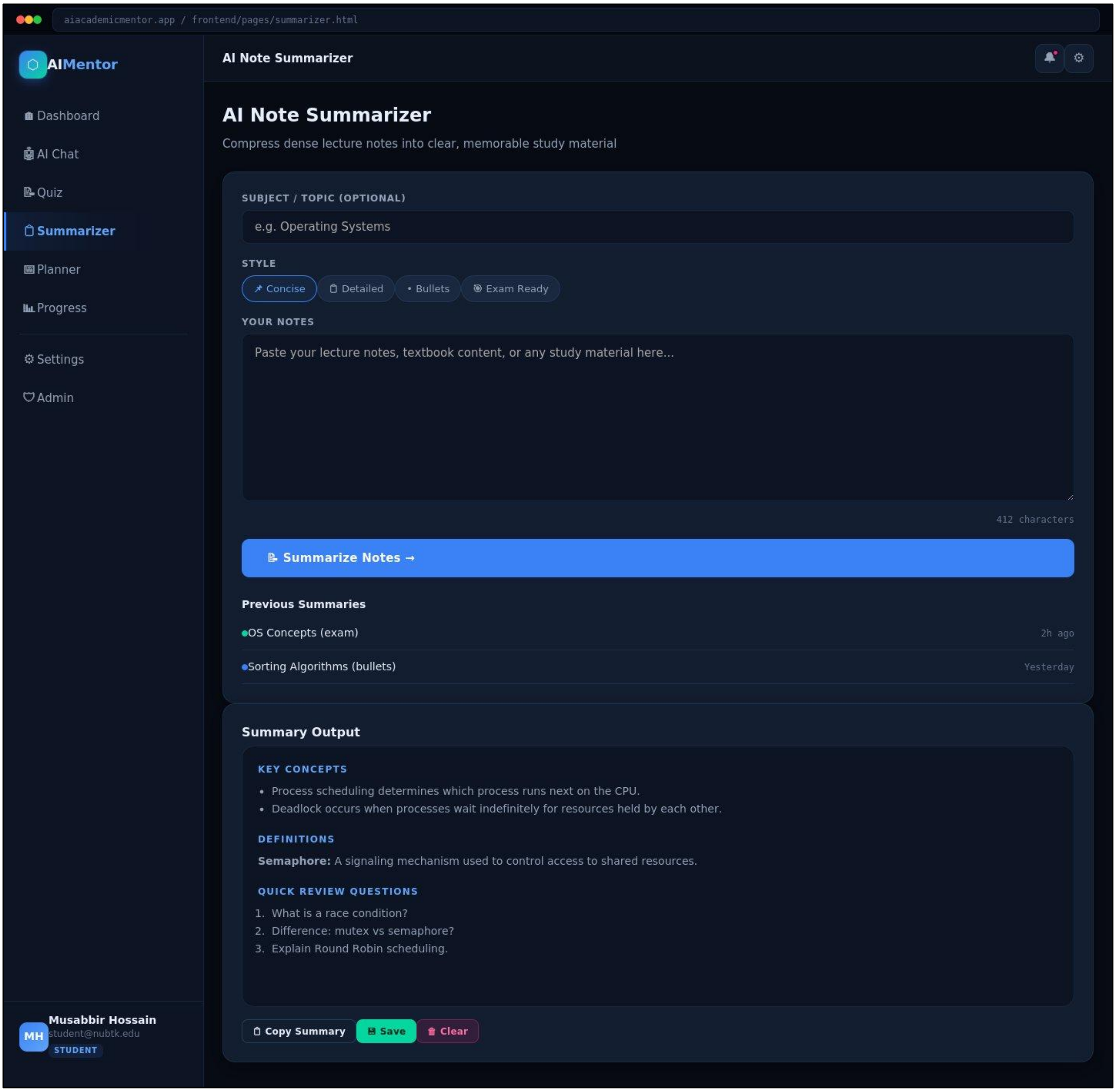

Figure: *Note Summarizer — Desktop View (1440px)*

Mobile Responsive View (390px):

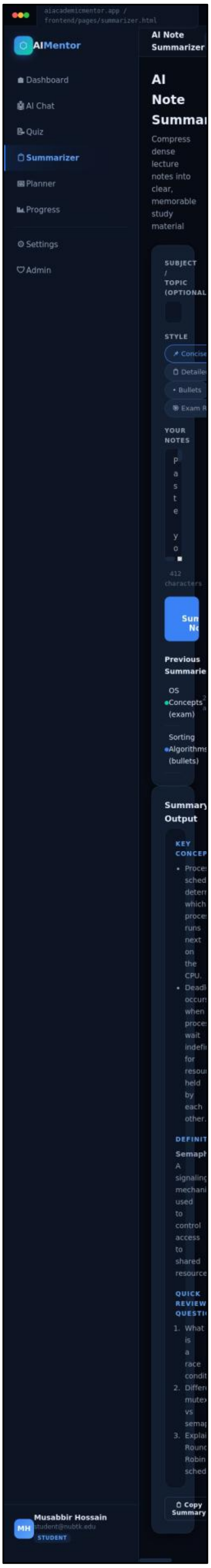

Figure: *Note Summarizer — Mobile View (390px)*

Key UI Elements:

- **Input Area:** Large resizable textarea for pasting raw lecture notes
- **File Upload Zone:** Drag-and-drop upload panel for PDF and text documents
- **Output Card:** AI-generated summary in formatted card with copy and download actions
- **Length Control:** Concise / Detailed / Bullet Points output format chips
- **Subject Tag:** Auto-detected subject label shown on generated summary card

Screen 8: Study Planner

Comprehensive study schedule management tool. Students create, rearrange, and track academic tasks across the week with deadline urgency indicators and per-subject color coding.

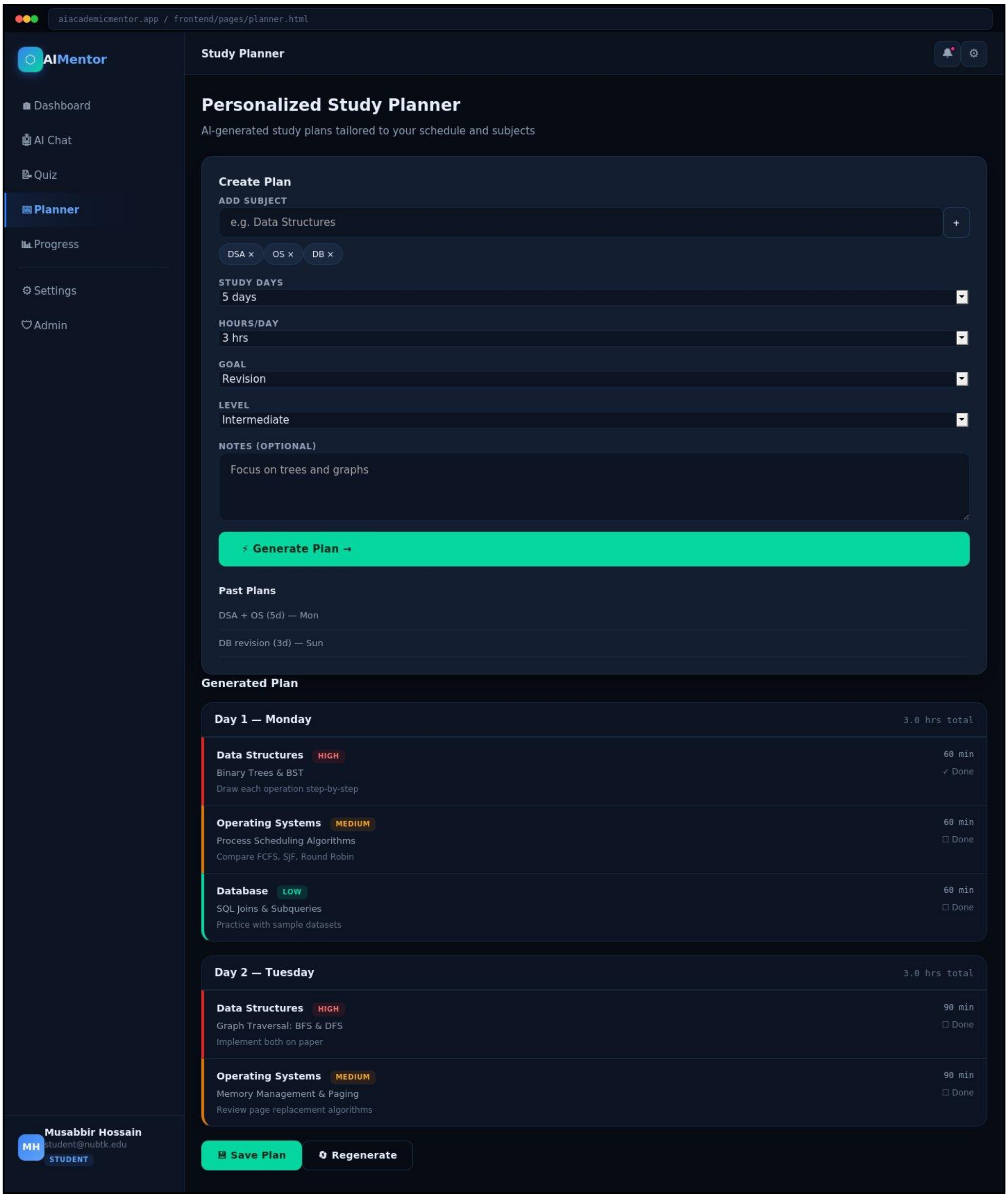

Figure: *Study Planner — Desktop View (1440px)*

Mobile Responsive View (390px):

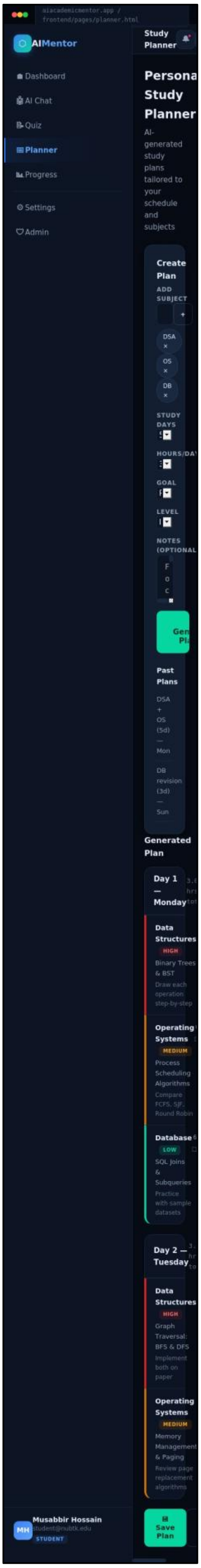

Figure: *Study Planner — Mobile View (390px)*

Key UI Elements:

- **Weekly Grid:** 7-column time-blocked weekly schedule with day headers and time slots
- **Task Cards:** Color-coded by subject with priority indicator and deadline label
- **Add Task Modal:** Inline form for subject, date, time, duration, and priority selection
- **Progress Chip:** Per-day completion percentage shown below day header
- **Deadline Alert:** Red-bordered urgency card for tasks due within 24 hours

Screen 9: Progress Analytics

Data-rich analytics view giving students full visibility into their learning trajectory. Topic mastery, streak data, and historical scores are visualized in an intuitive dashboard layout.

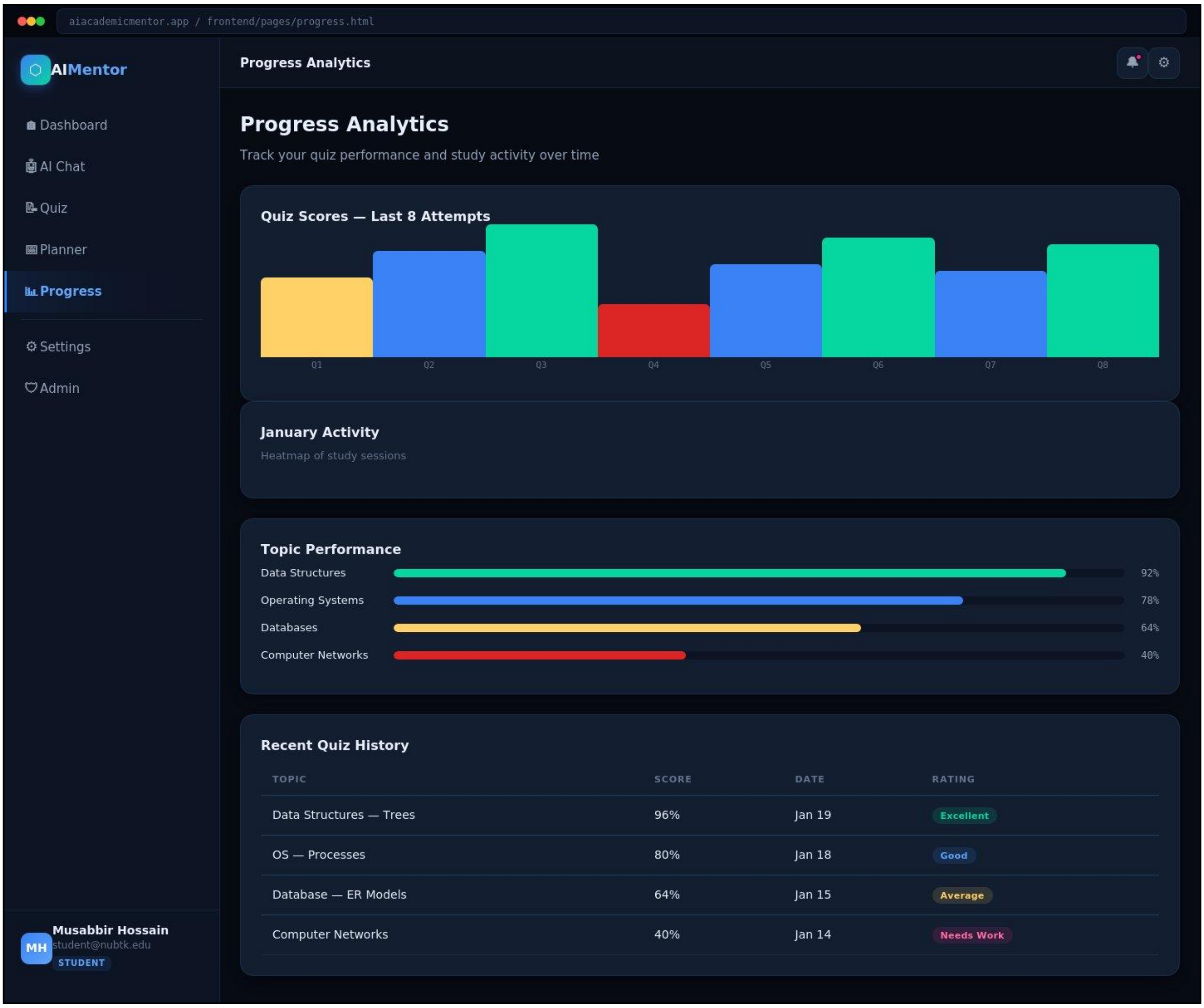

Figure: *Progress Analytics — Desktop View (1440px)*

Mobile Responsive View (390px):

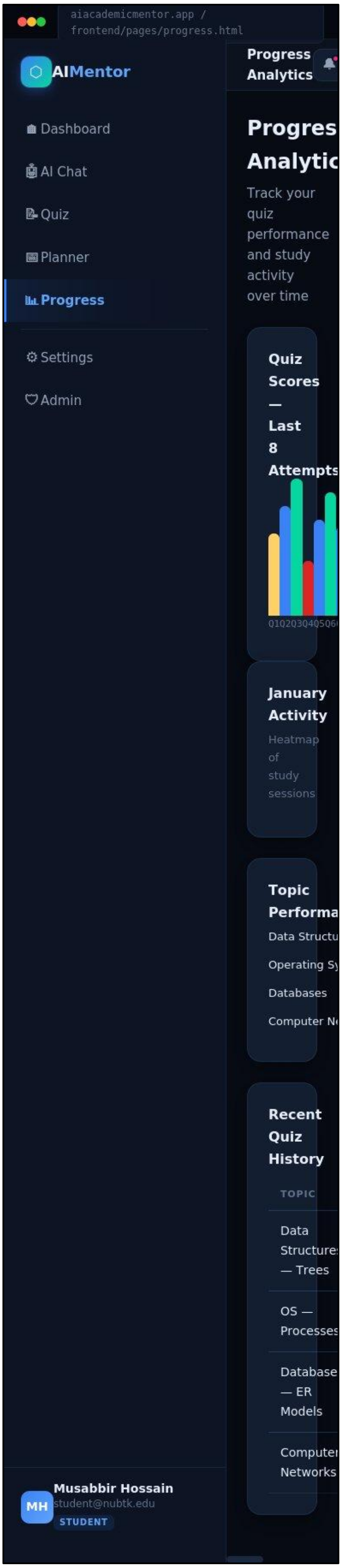

Figure: Progress Analytics — Mobile View (390px)

Key UI Elements:

- **Activity Heatmap:** 7-column grid with 4-level green intensity showing daily study activity
- **Topic Mastery Bars:** Per-subject horizontal bar chart with score percentage and color coding
- **Score Trend Chart:** Bar chart showing quiz performance over last 7 attempts by week
- **Streak Counter:** Prominent current streak display with longest streak record
- **Weak Area Flags:** AI-identified topics below 60% highlighted with improvement suggestions

Screen 10: Notifications

Centralized notification center with categorized academic alerts. Study reminders, quiz result notifications, and system messages are displayed in a clean time-sorted feed.

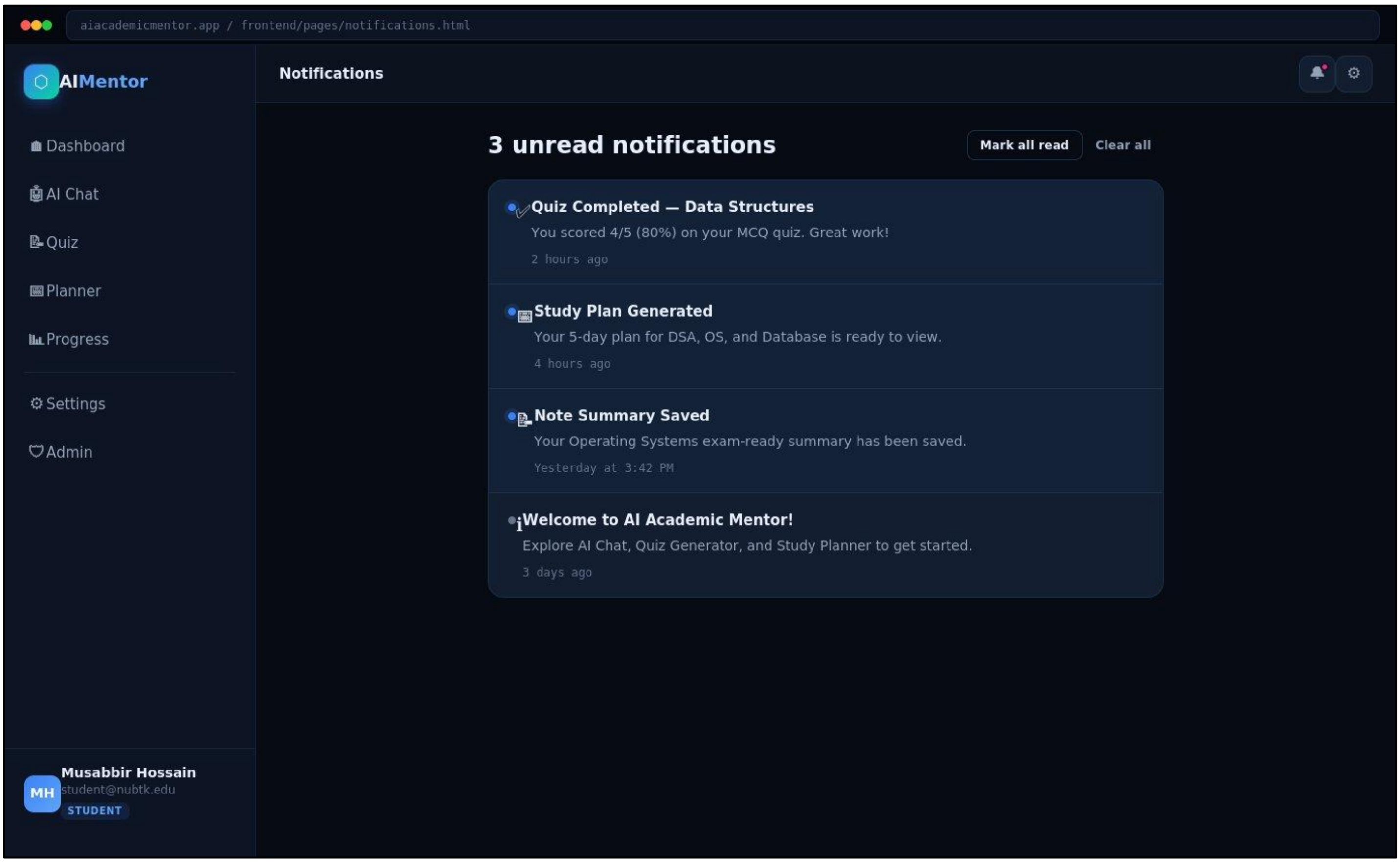

Figure: *Notifications — Desktop View (1440px)*

Mobile Responsive View (390px):

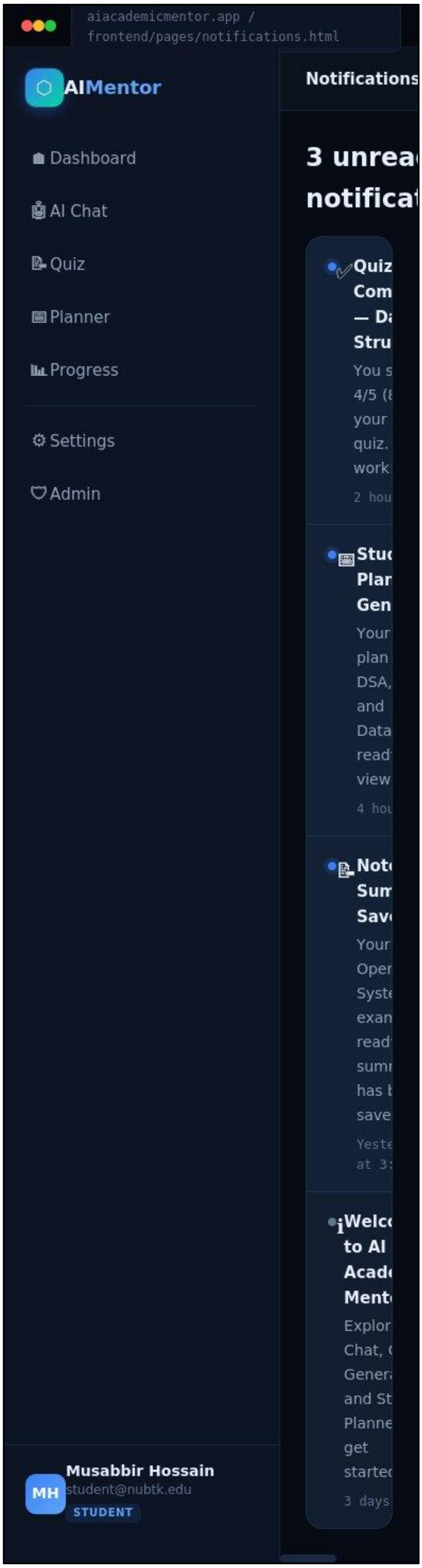

Figure: Notifications — Mobile View (390px)

Key UI Elements:

- **Notification Feed:** Time-ordered list with category icons: study reminders, quiz results, system alerts
- **Unread Badge:** Red dot-badge on notification bell icon in topbar
- **Filter Chips:** All / Unread / Quizzes / Planner / System chip row for category filtering
- **Mark All Read:** One-click bulk action button in topbar actions row
- **Timestamp:** JetBrains Mono relative timestamps (2h ago, Yesterday, etc.)

Screen 11: Settings & Profile

Two-column settings layout with profile information on the left and preferences on the right. The danger zone section is visually separated and requires explicit confirmation before destructive actions.

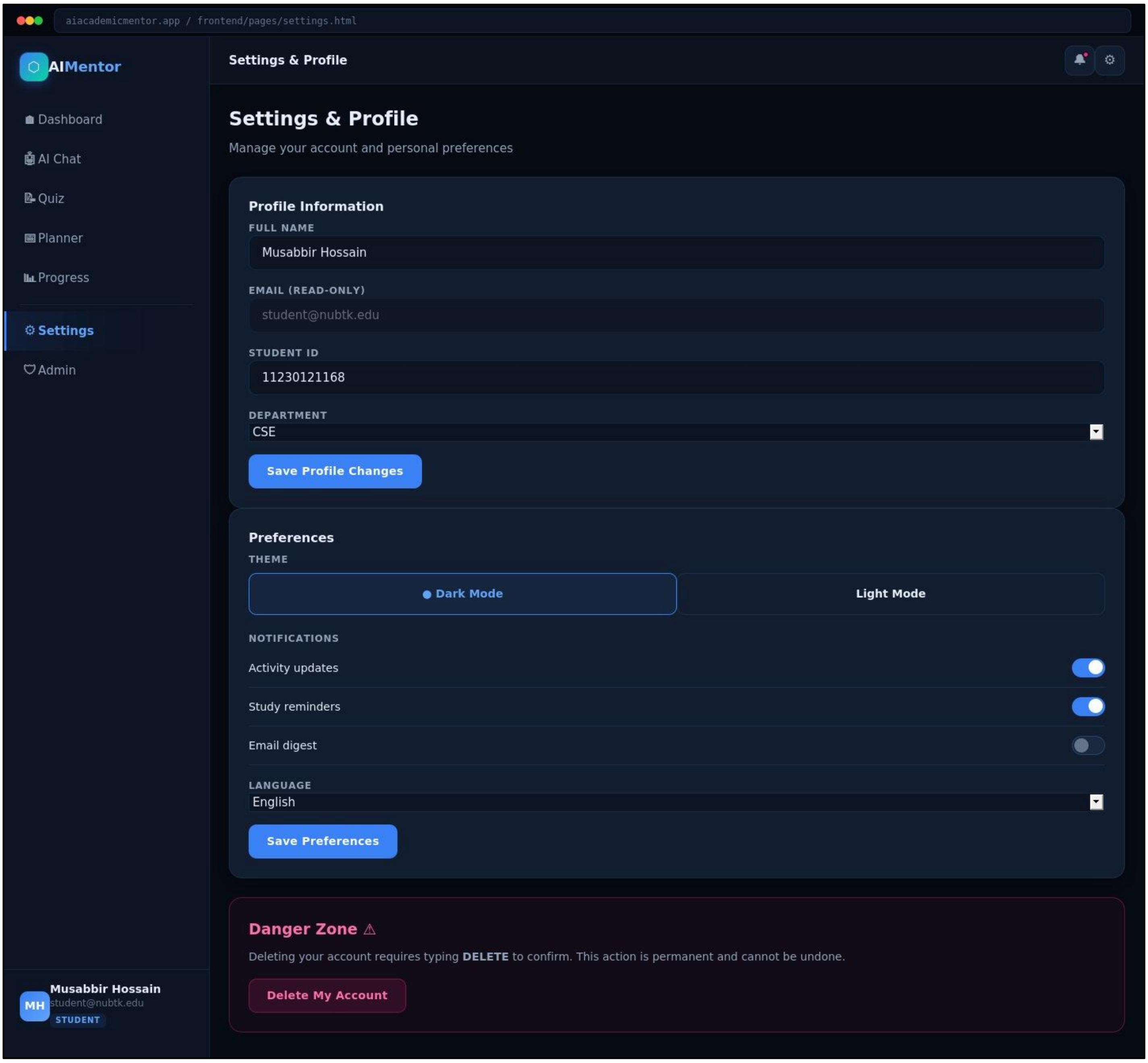

Figure: *Settings & Profile — Desktop View (1440px)*

Mobile Responsive View (390px):

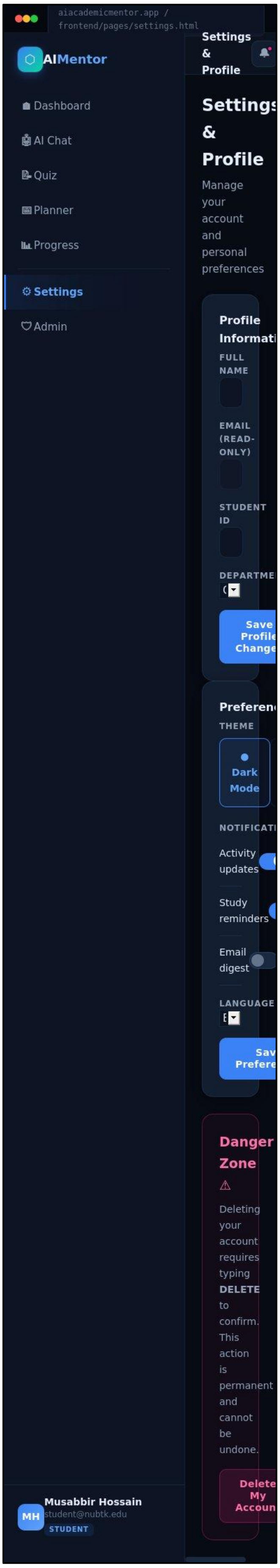

Figure: Settings & Profile — Mobile View (390px)

Key UI Elements:

- **Profile Form:** Full Name, Email (read-only), Student ID, and Department editable fields
- **Theme Toggle:** Dark / Light mode chip selector with active state styling
- **Notification Prefs:** Toggle switches for Activity Updates, Study Reminders, Email Digest
- **Language Select:** Dropdown for interface language selection
- **Danger Zone:** Red-bordered section with type-to-confirm account deletion CTA

Screen 12: Admin Panel

Restricted admin-only view accessible only to accounts with the admin role. The panel provides complete platform oversight including user lifecycle management, content announcements, and system configuration.

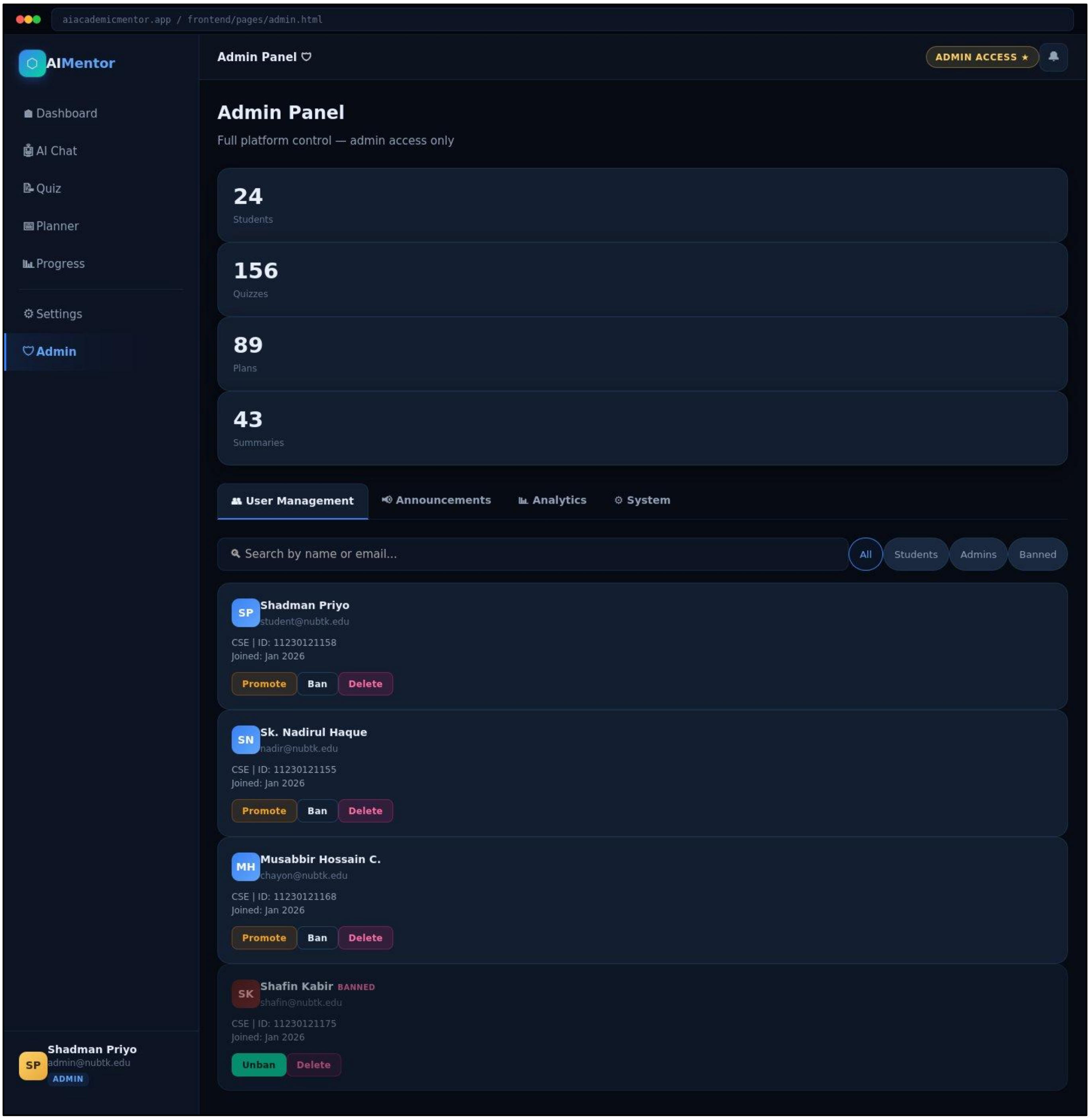

Figure: Admin Panel — Desktop View (1440px)

Mobile Responsive View (390px):

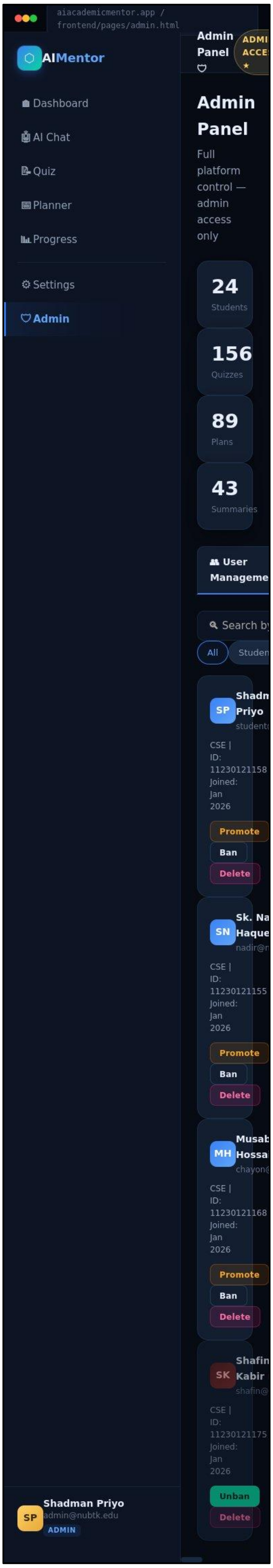

Figure: *Admin Panel — Mobile View (390px)*

Key UI Elements:

- **Admin Stats Row:** Four summary cards: total Students, Quizzes, Plans, Summaries
- **User Management Tab:** Card grid showing student profiles with Promote, Ban, and Delete actions
- **Banned User State:** Reduced opacity card with red "Banned" label and Unban/Delete actions only
- **Search & Filter:** Search input with All / Students / Admins / Banned filter chips
- **Admin Badge:** Gold "Admin Access ★" badge in topbar, gold avatar for admin user
- **Tab Navigation:** User Mgmt · Announcements · Analytics · System Config tabs

2. User Flow Design

Detailed Flow Table

Step	Screen	User Action / System Response	Access Level
1	**Landing Page (/)**	User visits the application at the root URL; sees landing page with feature overview and CTA buttons	Public (no auth required)
2	**Register (/register)**	New user fills registration form: First Name, Last Name, Email, Student ID, Department, Password, Role (Student/Admin)	Public (no auth required)
3	**Login (/login)**	Returning user authenticates via email + password; system issues JWT token on success	Public (no auth required)
4	**Dashboard (/dashboard)**	Post-login home: user views academic metrics (quizzes, streak, score), quick-access cards, and upcoming tasks	Authenticated (Student / Admin)
5	**Feature Access**	User selects a feature from the sidebar: AI Chat, Quiz Generator, Note Summarizer, or Study Planner	Authenticated (Student)
6	**AI Interaction**	User interacts with the selected AI-powered feature; Gemini API processes the request and returns a response	Authenticated (Student)
7	**Results Display**	AI result rendered: chat response, quiz score card, structured summary, or study plan — all saved to MongoDB	Authenticated (Student)
8	**Progress Review (/progress)**	User reviews performance analytics: topic mastery bars, streak calendar, score trend chart, weak area flags	Authenticated (Student)
9	**Notifications (/notifications)**	User reviews academic reminders, quiz result alerts, and system messages in the notification center	Authenticated (Student)
10	**Settings (/settings)**	Optional: user updates profile, theme preference, notification toggles, or language setting	Authenticated (Student)
11	**Admin Panel (/admin)**	Admin-only: manage users (promote/ban/delete), post announcements, view system analytics, configure platform	Authenticated (Admin role only)

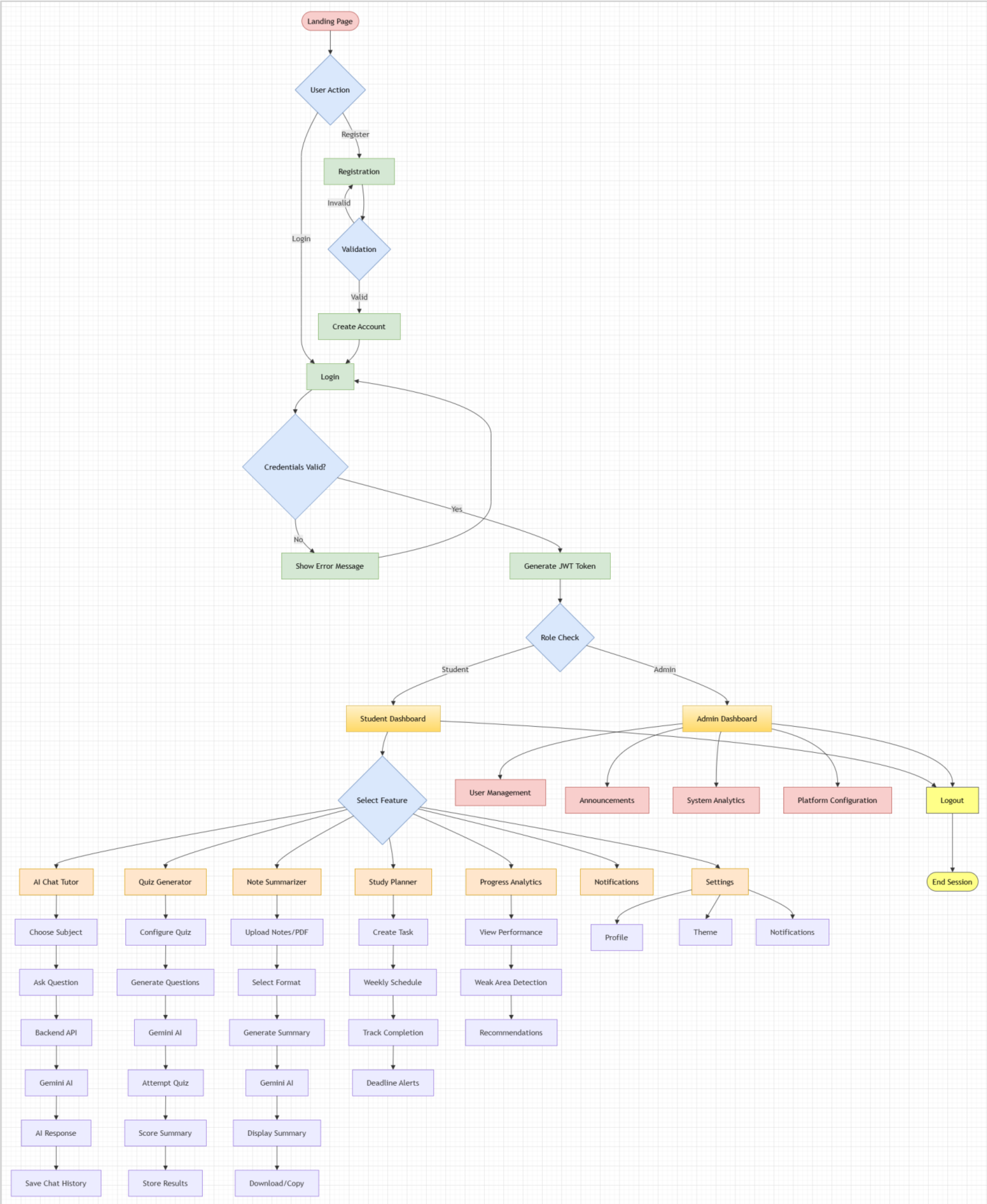

***Figure:* User Flow Design**

3. Development Roadmap

The AI Academic Mentor project follows a 14-week development timeline structured across six phases. Each phase has defined deliverables, key tasks, assigned team members, and a completion status. The roadmap below serves as the primary reference during implementation and evaluation.

Week	Phase	Status	Deliverables	Key Tasks
Week 1–2	**Planning**	**Completed**	GitHub Repository, Project Proposal PDF	— Team formation and role assignment (Leader, Frontend, Backend, AI Integration Lead) — GitHub repo created with branch structure: main / development / frontend / backend / ai-integration — Technology stack finalized: React.js, Node.js, MongoDB, Gemini API — Week 2 Project Proposal document submitted
Week 3–5	**Design**	**Completed**	SRS Document, ER Diagram, Architecture Diagram, UI/UX Wireframes	— Week 3: Software Requirements Specification (SRS) + use case diagrams — Week 4: System architecture diagram, ER diagram, and REST API structure — Week 5: Figma wireframes for all 12 major screens — Week 5: UI Screenshots document and Development Planning document
Week 6–7	**Development**	**Upcoming**	Auth APIs, Database Schemas, All UI Pages Connected to Backend	— Week 6: JWT auth system, bcrypt password hashing, REST API endpoints — Week 6: MongoDB Atlas setup and database schemas for users, quizzes, study plans, progress — Week 7: Build all 12 UI pages in React.js + Tailwind CSS — Week 7: Connect all React pages to backend REST APIs and test data flow
Week 8–9	**AI Integration**	**Upcoming**	AI Chatbot, Quiz Generator, Note Summarizer, Study Planner, Dashboard	— Week 8: Integrate AI Chatbot (Gemini API) with conversation history per session — Week 8: Build AI Quiz Generator — MCQ + short-answer formats, score tracking in MongoDB — Week 8: Implement AI Note Summarizer — paste or upload notes, generate summaries — Week 9: Build AI-powered Personalized Study Planner with structured day-by-day plans

				– Week 9: Complete Progress Dashboard with charts, topic-wise analytics, and engagement metrics – Week 9: Polish UI, add in-app notification system and Admin Panel
Week 10–11	**QA & Deploy**	**Upcoming**	Testing Report, Live Deployment (Vercel + Render/Railway)	– Week 10: Write test cases; run unit, integration, and user acceptance testing – Week 10: Document all bugs and fixes in a testing report – Week 11: Deploy frontend to Vercel – Week 11: Deploy backend API to Render / Railway – Week 11: Verify live deployment and update README.md with deployment link
Week 12–14	**Presentation**	**Upcoming**	Final Report, User Manual, API Documentation, Live Demo	– Week 12: Write final project report, user manual, and API documentation – Week 13: Prepare presentation slides and rehearse live demonstration – Week 14: Final presentation, live demo, Q&A, and viva evaluation

4. Team Task Distribution

Each team member has clearly defined responsibilities aligned to their technical role. The distribution ensures full coverage of all system components with no single point of failure. Every member is accountable for a distinct set of deliverables across the development lifecycle.

Member	Student ID	Role	Responsibilities
Shadman Sadequeen Priyo	11230121158	**Team Leader**	– Project coordination and milestone management – GitHub repository management and branch strategy – System architecture oversight and design review – Final documentation compilation and submission – Communication with supervisor (Md. Riaz Mahmud)
Musabbir Hossain Chayon	11230121168	**Frontend Developer**	– React.js + Tailwind CSS UI implementation – All 12 application page screens – Figma wireframes and UI/UX design system

			— API integration: connecting frontend to backend REST endpoints — Responsive design implementation (desktop 1440px + mobile 390px) — Frontend deployment to Vercel
Sk. Nadirul Haque Nadir	11230121155	**Backend Developer**	— Node.js + Express.js REST API development — MongoDB Atlas database setup and schema design — JWT authentication system and bcrypt password hashing — All database models: users, quizzes, plans, progress — API documentation and backend deployment to Render / Railway
Shafin Kabir	11230121175	**AI Integration Lead**	— Gemini API integration and prompt engineering — AI Chatbot tutor module development — AI Quiz Generator (MCQ + short-answer) implementation — AI Note Summarizer module development — Personalized AI Study Planner feature — AI module testing and optimization

5. Final Technology Stack

The following technology stack has been confirmed and finalized for the AI Academic Mentor project. All technology choices were made to prioritize developer productivity, scalability, and alignment with the team's existing skill set.

Layer	Technology / Tool	Purpose
Frontend	**React.js + Tailwind CSS**	User interface and responsive design
State Management	**React Context API / useState / useEffect**	Client-side state, session handling, and side effects
Backend	**Node.js + Express.js**	REST API, business logic, authentication
Database	**MongoDB + MongoDB Atlas**	Flexible NoSQL data storage and cloud hosting
AI Integration	**Gemini API (Google)**	Chatbot tutoring, quiz generation, note summarization, study planning
Authentication	**JWT + bcrypt**	Secure login, session tokens, and password hashing
UI Design Tool	**Figma**	High-fidelity wireframes and interactive prototype
Design System	**Custom HTML/CSS + Sora + JetBrains Mono**	Premium dark-themed UI component library
Frontend Deploy	**Vercel**	Fast, scalable frontend hosting with CI/CD

Backend Deploy	**Render / Railway**	Free-tier backend API hosting
Version Control	**Git + GitHub**	Code management and team collaboration
Icon Set	**System Emoji Icons**	Universally compatible, no external dependency

5.1 Technology Justification

React.js + Tailwind CSS

Chosen for component reusability, virtual DOM performance, and a large ecosystem. Tailwind CSS enables rapid UI development with utility-first styling consistent with the design token system.

Node.js + Express.js

JavaScript across the full stack simplifies context-switching for the team. Express.js provides a minimal, unopinionated REST API framework that scales cleanly with the project's endpoint count.

MongoDB + MongoDB Atlas

The flexible document schema suits the heterogeneous data structures in the application (user profiles, quiz sessions, study plans, chat history). MongoDB Atlas provides cloud hosting with a generous free tier.

Gemini API (Google)

Selected for its generous free tier, strong contextual understanding, and proven performance on academic content generation tasks including MCQ creation, text summarization, and study plan generation.

Vercel + Render / Railway (Deployment)

Zero-cost deployment platforms with CI/CD integration to GitHub. Vercel specializes in React/Next.js frontends; Render/Railway provides reliable free-tier Node.js hosting with automatic deploys.